



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SALEM DISTRICT LORRY OWNERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

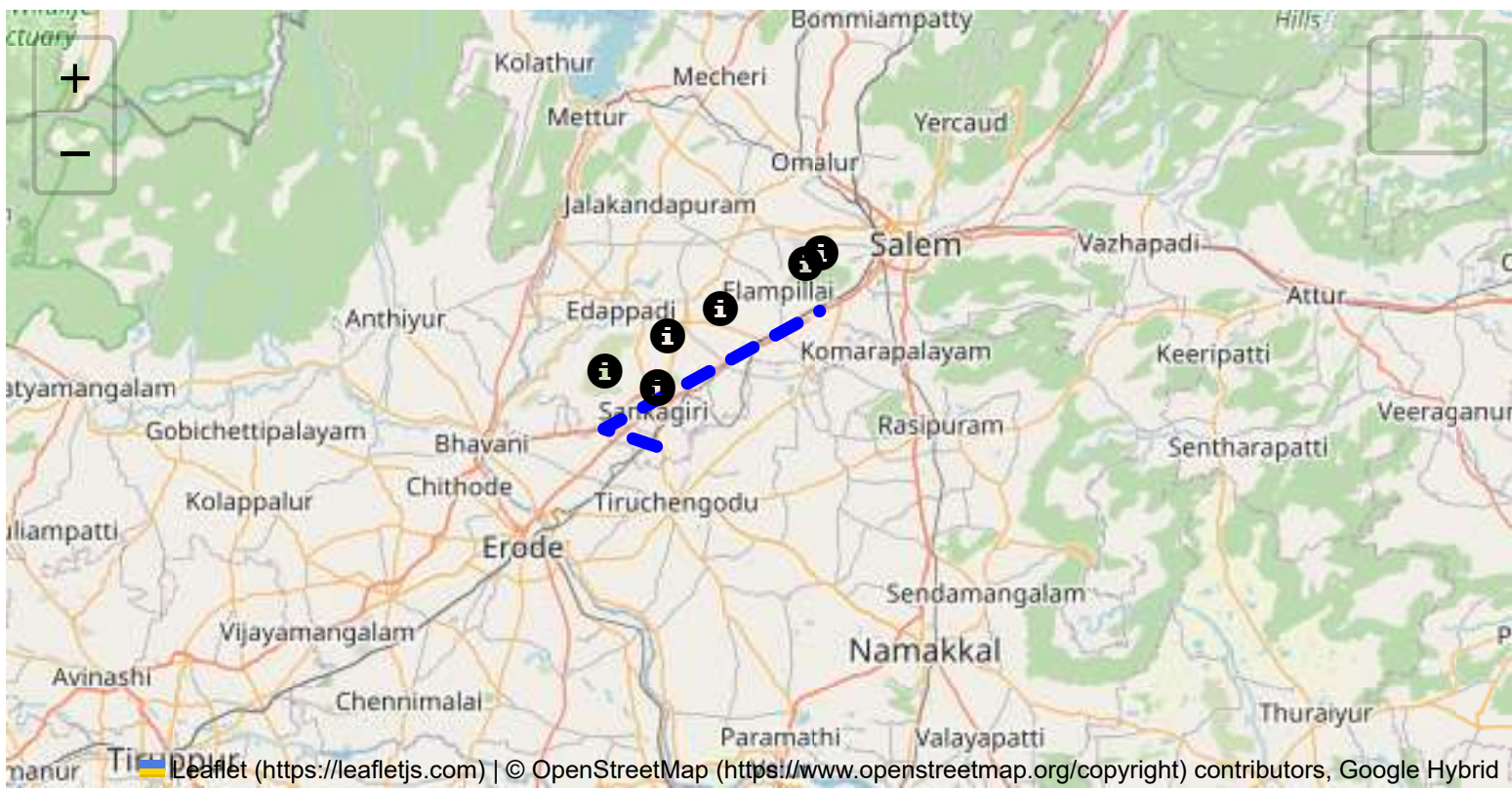
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

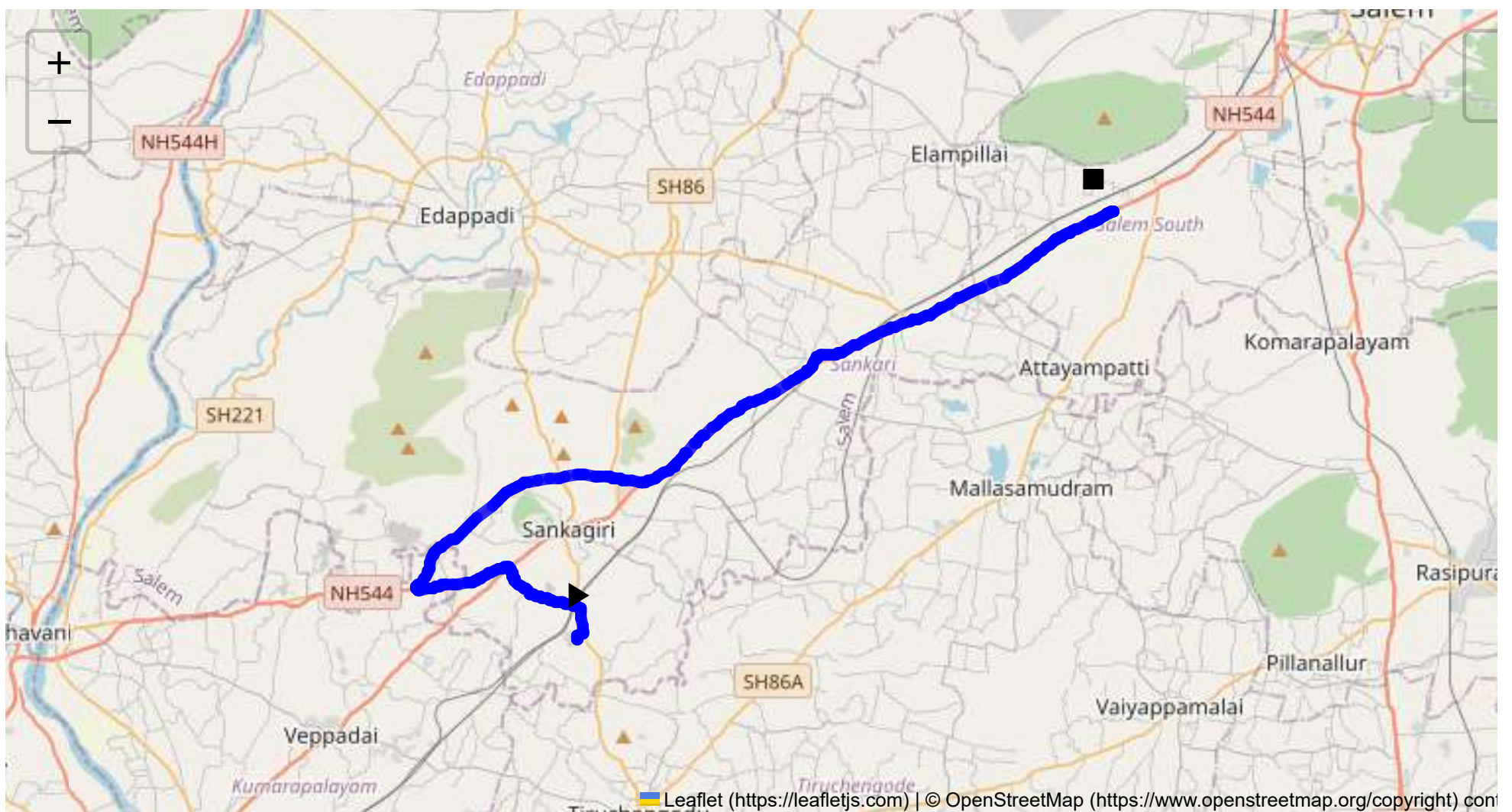
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 40.98 km
Estimated Duration: 0.8 hours
Adjusted Duration (Heavy Vehicle): 1.0 hours
Start: (11.4381, 77.8734)
End: (11.581229, 78.053291)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route runs through various towns and rural areas within Tamil Nadu, covering approximately 41 kilometers starting from Sangagiri and ending in Chinnasiragapadi, Salem. It passes through Puthur, Padaiveedu, Sangagiri, Vaikundam, and Salem. The path mostly follows major highways and service roads, indicating a relatively straightforward route suitable for heavy vehicles.

2. Weather Conditions and Potential Weather-related Hazards: Tamil Nadu experiences a tropical climate with a distinct monsoon season. The route may face moderate to heavy rainfall from October to December due to the northeast monsoon, potentially causing waterlogging. Extreme heat can be expected from March to June. Drivers should be aware of reduced visibility and slippery conditions during the monsoon.

3. Analysis of Traffic Patterns: Traffic congestion can be expected in urban areas and market zones, especially during peak hours between 8-10 AM and 5-8 PM. Highways can witness steady traffic flow, but junctions and service roads near Sangagiri and Salem might be prone to bottlenecks.

4. Assessment of Road Quality and Infrastructure: The highway sections are generally well-maintained, but some rural roads may have potholes or uneven surfaces. Pay attention to construction zones or roadwork which are fairly common and may reduce road quality temporarily.

5. Suggestions for Alternative Routes: In case of an emergency, detours can be made via smaller connecting roads leading back to the NH 544, or by using local roads that connect to nearby towns with medical or support facilities. It's important to possess updated maps or GPS with live traffic updates.

6. Local Regulations Affecting Hazardous Material Transport: Tamil Nadu follows national guidelines for the transport of hazardous materials, including restrictions on certain routes during peak traffic hours. Compliance with vehicle safety and signage is critical.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials: There have been occasional incidents involving heavy vehicles, primarily due to brake failures or abrupt lane changes on highways. Historically, mishandling of hazardous materials has led to spills, often due to inadequate containment.

8. Environmental Considerations and Sensitive Areas: The route does not pass through any forest reserves or environmentally sensitive zones but caution is advised near agricultural lands to prevent any contamination. Measures should be taken to ensure no spillage affects soil or water bodies.

9. Analysis of Communication Coverage: Most of the route enjoys good mobile network connectivity, but certain rural segments might experience spotty coverage. Emergency contact systems should be tested in these areas.

10. Estimated Emergency Response Times: Emergency services are usually accessible within 15-30 minutes in urban segments of the route, such as near Sangagiri and Salem, but it could extend to 45 minutes in more rural parts. Familiarity with local emergency contact numbers is crucial.

11. Overall Summary of Risk Assessment: The route is considered moderately safe for heavy vehicle transport with adequate infrastructure and services. However, it requires vigilance regarding traffic patterns, especially during peak hours and monsoon seasons. Compliance with regulations is critical to avoid penalties and ensure safety. Preparing for potential road quality issues in rural sections and understanding emergency procedures ahead of time will contribute to a safer journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43961, 77.87341	30 KM/Hr
2	Turn	Medium	11.43968, 77.87345	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44898, 77.87410	30 KM/Hr
5	Turn	Medium	11.45357, 77.85789	30 KM/Hr
6	Turn	Medium	11.45348, 77.85706	30 KM/Hr
7	Turn	Medium	11.45352, 77.85698	30 KM/Hr
8	Turn	Medium	11.46318, 77.84913	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Turn	High	11.45631, 77.81668	15 KM/Hr
11	Turn	High	11.49409, 77.88004	15 KM/Hr
12	Turn	Medium	11.49419, 77.88015	30 KM/Hr
13	Turn	Medium	11.49377, 77.88573	30 KM/Hr
14	Turn	High	11.49363, 77.88590	15 KM/Hr
15	Turn	High	11.49235, 77.89561	15 KM/Hr
16	Blind Spot	Blind Spot	11.49225, 77.89560	10 KM/Hr
17	Blind Spot	Blind Spot	11.58526, 78.06049	10 KM/Hr
18	Blind Spot	Blind Spot	11.58499, 78.06061	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45348, 77.85706



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49409, 77.88004



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88015



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49377, 77.88573



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49363, 77.88590



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49235, 77.89561



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49225, 77.89560



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.58526, 78.06049



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.58499, 78.06061

